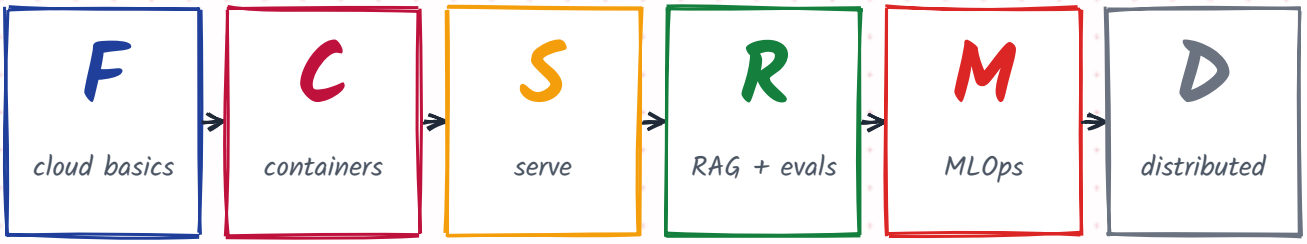
Your Learning Path

Lesson 12 of 12 · From zero to cloud AI engineer, in 16 weeks.

In this lesson → ① Six stops ② Projects and done-when checks ③ Time and money ④ The forgotten GPU ⑤ Certificates in 2026 ⑥ What employers ask ⑦ The claims ledger
Course project: ShopNest Copilot, the AI support assistant we moved to the cloud, step by step.

1 The road: six stops



Frugal Coders Spend Really Modest Dollars

> **F-C-S-R-M-D** = "Frugal Coders Spend Really Modest Dollars": Fundamentals, Containers, Serving, RAG, MLOps, Distributed. Each stop uses the one before. The order is my synthesis (job postings + cost traps), not an official ranking.

- → Run stop 4 (RAG, agents, evals) alongside stops 2-3: almost no cloud spend, and foundation models appear in 51% of postings
- → Every stop ends with a project, an estimated cost and a **done when** check
- → Do not skip the check: being able to show the thing working turns a course into portfolio evidence

2 Stops 1 and 2

Stop	Learn	Project	Done when
1 Cloud fundamentals (wk 1-2)	IAM, private network, object storage, GPU quotas, budgets	Terraform: network + least-privilege role + encrypted bucket + budget alert	destroy then apply rebuilds it, no console clicks · other bucket DENIED · budget alert fired
2 Containers + K8s (wk 3-5)	Dockerfiles for ML, registries, Deployment, autoscaling, GPU node pools	containerised inference service on managed K8s; Terraform GPU pool, one L4, scale to zero	node shows nvidia.com/gpu: 1 · pod runs nvidia-smi · pool back to 0 nodes when idle

Request GPU quota in week 1: new accounts may need it (unverified for 2026).

3 Stops 3 and 4

Stop	Learn	Project	Done when
3 Serving (wk 6-7)	vLLM, SGLang, TGI, Triton, Ray Serve; TTFT, tokens/s, p95	vLLM serving a 7-8B model on one L4 / A10G / L40S; benchmark	table of tokens/s, TTFT and p95 at 3 concurrency levels x 2 precisions · explain why throughput and p95 both rise
4 RAG, agents, MCP + evals (wk 8-10)	RAG pipeline, agents, tool calling, MCP, tracing, recall + faithfulness evals	RAG app + eval suite in CI; MCP server + traced agent	CI red when recall drops · every tool call traced · eval set size and gaps stated

```
# Stage 3 lab: ~7.6B params x 2 B (BF16) = ~15 GB, fits a 24 GB card
vllm serve Qwen/Qwen2.5-7B-Instruct --max-model-len 8192 --gpu-memory-utilization 0.90
for c in 1 8 32; do
  vllm bench serve --model Qwen/Qwen2.5-7B-Instruct --dataset-name random \
    --random-input-len 512 --random-output-len 128 --num-prompts 200 --max-concurrency $c
done
```

4 Stops 5 and 6

Stop	Learn	Project	Done when
5 MLOps + observability (wk 11-13)	tracking, registry, CI/CD with eval gate, canary, Terraform environments, drift	LoRA on spot, checkpoints to a bucket, register, serve with vLLM; CI/CD + drift dashboard	kill the spot VM, job resumes, loss curve continuous · versioned model with metrics · alert fires on fake drift
6 Distributed training (wk 14-16)	DP / TP / PP, ZeRO + FSDP, collectives, MFU and memory accounting	DDP on free 2xT4, layouts on 2-8 rented GPUs, nanochat speedrun (~2 h on 8xH100, about $48, $15 spot)	say from model size whether DDP fits · measured memory roughly agrees · MFU table · chat CLI works

Spot cost model: spot $ = p_spot x (H + k x h_lost). Without checkpoints, h_lost grows with H (lesson 2).

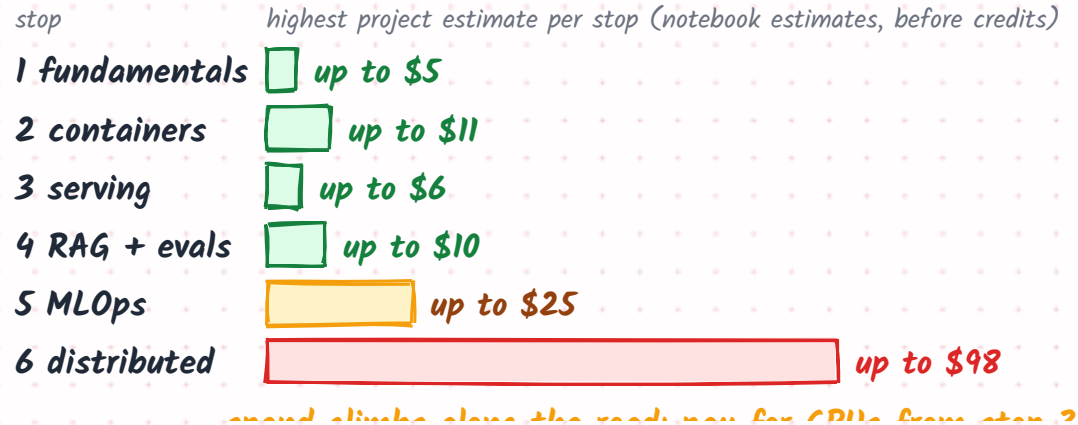
5 How long, how much

Arithmetic. 16 weeks x about 10 h = **160 hours** (the 10 is my assumption). Script: l12_study_plan.py.



12-week fast track: skip or merge the four starred weeks (quantisation, agents + MCP, observability, efficiency).

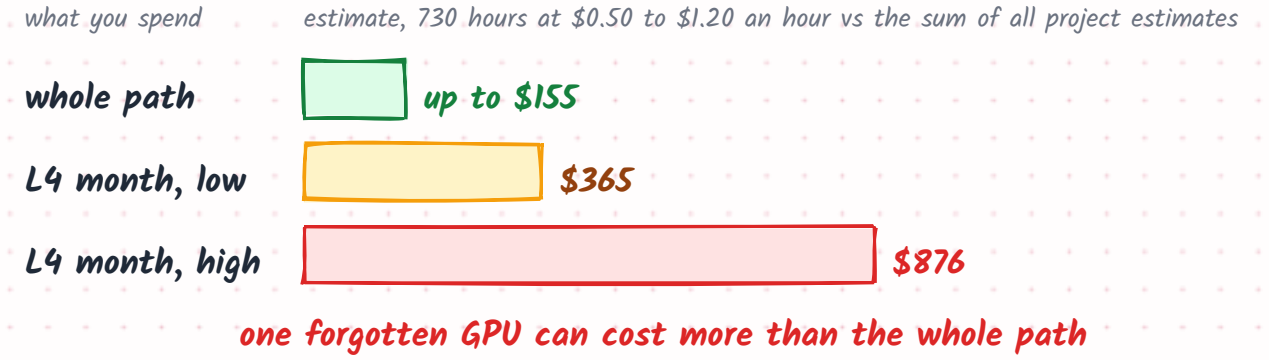
6 What the path costs



	Estimate
sum of project estimates (my addition)	$32 to $155 before credits
stops 1–5 only	$7 to $57
notebook estimate, whole path	**about $150–250** out of pocket, if you tear down
new-account credits (2-3-2)	AWS $200 / 12 months · GCP $300 / 90 days · Azure $200 / 30 days
free GPU notebooks	Kaggle about 30 GPU-hours a week

Budget the higher figure. Spend credits on CPU, networking and Kubernetes stops; use free notebooks for learning-size GPU work; rent marketplace GPUs for stops 3, 5 and 6.

7 The forgotten GPU



one forgotten GPU can cost more than the whole path

One L4 node left up for a month: 730 h x $0.50-1.20 = **$365 to $876**: 2.4 to 5.7 times the top-end cost of the whole path. Idle resources are the number one way to overspend.

```
# after EVERY lab: prove nothing is still billing
terraform destroy -auto-approve
gcloud container clusters list       # expect: empty
aws eks list-clusters --output text  # expect: empty
```

- ✓ Every Friday: run the teardown commands and check the billing page
- ✓ NAT gateways in "free" networking labs keep billing by the hour: avoid or destroy the same day
- ✓ EKS: $0.10/h standard vs about $0.60/h on extended support: keep versions current
- ✓ GKE $74.40 credit: zonal or Autopilot only, never nodes
- ✓ Spot without checkpoints can cost more than on-demand

8 Cheap GPUs by stop

GPU	About $/GPU-hr (Sep 2026, vendor and aggregator)	Use
T4 / P100	free on Kaggle / Colab	stop 1, DDP basics
L4 / A10G (24 GB)	$0.50-1.50	stops 2-3
L40S (48 GB)	$0.92-0.99	stop 3 benchmarks
A100 80 GB	$1.49 (RunPod spot from about $0.36)	stop 5 LoRA
H100	$1.33-3.00 on marketplaces; AWS $6.88	stop 6 nanochat

These are vendor and aggregator figures, not provider price lists. No reliable 2026 spot data was found for L4 / A10G.

9 Certificates in 2026: check the code

Retired or changed	What happened	Next
AWS ML Specialty MLS-C01	retired, last exam Mar 31, 2026; holders stay active 3 years from date earned	AIP-C01 or MLA-C02
AWS MLA-C01	English C01 last day Sept 28, 2026	MLA-C02 (same credential name)
Azure AI-102	retired Jun 30, 2026	AI-103 (a new exam)
Azure DP-100	retired Jun 1, 2026	AI-300 MLOps Engineer
Azure AI-900	new version Jun 30, 2026	AI-901

Who went MAD: MLS-C01, AI-102, DP-100. Check the exam code on the official page right before booking.

10 Which cert, when

Stop	Cert (cost)	Note
1	AWS AI Practitioner AIF-C01 ($100)	foundational
2	CNCF KCNA ($250), then CKA / CKAD ($445)	CKA and CKAD are hands-on, in a live terminal
4	AWS AIP-C01 ($300) or Azure AI-103	generative AI applications and agents
5	Google PMLE ($200) or AWS MLA-C02 or CNCF CNPE ($445)	MLOps, serving, monitoring
6	NVIDIA NCP-GENL ($200)	LLM professional

Hands-on exams are probably a stronger signal than multiple choice: inferred from format, not from an employer survey. No data shows how often postings require a specific cert: **projects beat badges**. Prices vary by source: confirm before booking.

11 Books and courses: a minimal stack

- → Short on time: Karpathy **Zero to Hero** → **Made With ML** → Huyen **AI Engineering** + HF **Agents** course → **Ultra-Scale Playbook** + **nanochat**
- → Free: fast.ai, HF LLM and MCP courses, Stanford CS336 lectures, Full Stack Deep Learning (2022, tools dated)
- → Huyen **Designing ML Systems** covers classic ML systems; AI Engineering covers foundation-model apps
- → Provider paths (AWS Skill Builder, Google Cloud Skills Boost, Microsoft Learn): the standard free route; catalogs not rechecked

12 What employers ask for

Skill (43,480 US AI-engineering postings to Jul 2026)	Share
Python	62%
cloud platforms	55%
foundation models	51%

One analyst sample; about 1,550 new US roles a week. Platform / LLMOps roles add vLLM, Kubernetes + GPUs, Terraform, observability and usually want 3 to 7 years. Switching careers: aim at an applied AI engineer role (stops 1-4), add platform skills on the job.

13 Name projects in the posting's words

- ✓ "vLLM throughput benchmark"
- ✓ "Terraform GPU node pool"
- ✓ "RAG eval suite in CI"
- ✓ "LoRA on spot with checkpointing"

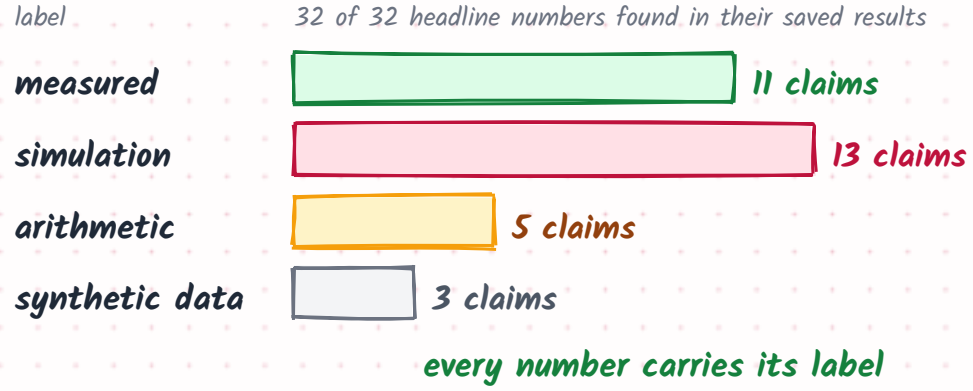
Put those phrases in your repository README files and résumé. Show the number you measured, what it cost, and what it does not cover.

14 Double-check before you pay

- → Prices, exam codes and credit rules change monthly: every dollar figure is a dated estimate
- → Sources disagree on several certificate prices and dates
- → Free-trial GPU quota was not rechecked in 2026: do not plan stop 3 around it
- → Posting samples are directional, not guarantees

15 The claims ledger

For 32 headline numbers from lessons 1-11, l12_claims_ledger.py looks the number up in the saved output of the lab that produced it and prints its label. **32 of 32 found.**



It proves each claim came from a file you can rerun, not that the experiment is perfect.

16 What the labs showed

Lesson	Headline	Label
1	70B FP8 + 32 chats of 8,192 tokens + headroom = 187 GB	arithmetic
2	no checkpoints on spot: 2,203,564 h for a 100 h job	simulation
4	20,000 small files: 25x slower than one shard	measured
5	Adam training state: 16.0 bytes per parameter	measured
6	KV cache: 12.1x faster generation	measured
7	eval gate blocked a prompt that raised the average	synthetic
10	filter 3 of 12 stopped; allowlist 0 leaks	simulation
11	timestamp on top of a prompt: 0% cached vs 74.9%	measured

17 ShopNest Copilot: the whole plan

Layer	Decision
capacity	fit memory first · spot with checkpoints · scale to zero
ship + data	layered images · sharded data · eval gate + canary
app	tested chunks · idempotent agent · egress allowlist · tenant filter
money	cache first · cost per resolved ticket · budgets and alerts

18 Your first week

- ✓ Open one cloud account and set a budget alert first
- ✓ Request GPU quota now
- ✓ Build the stop 1 Terraform project
- ✓ Run the lab scripts and change one number in each
- ✓ Friday: teardown commands and a billing check

19 Words to keep

Word	Meaning
Teardown	destroy what you built, then confirm nothing is still billing
Done when	a check that proves a stage works
Portfolio	a project you can show running, with its numbers
Label	is the number measured, simulated, calculated or made up?

20 Interview corner

Q. Why run RAG and agents alongside the infrastructure stops?

A. They need almost no cloud spend (about $0-10, mostly API tokens), and foundation models appear in 51% of postings.

Q. How can a spot run cost more than on-demand?

A. With no checkpoints, an eviction loses all work: cost is about p_spot x (H + k x h_lost). Checkpoint to object storage and resume.

Q. How do you avoid a surprise cloud bill?

A. Budget alert first, tear down after every lab, check for idle resources every week. One forgotten L4 for a month is $365 to $876.

21 Course complete

112_study_plan.py · 112_claims_ledger.py: change the hours, change a price, rerun.

⚡ Revise in 60 seconds

- → 6 stops: F-C-S-R-M-D, run stop 4 alongside
- → 16 weeks at about 10 h a week
- → about **$150–250** if you tear down
- → one forgotten GPU: **$365–876** a month
- → check the exam code before booking
- → projects beat badges
- → name projects in the posting's words
- → label every number you quote

Course complete
Thank you for learning with me. Go and build your first project this week.

Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**